

Rahul Lotlikar

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EDUCATION

Master of Science in Computer Science – University at Buffalo, SUNY Buffalo, NY

Dec 2025

Bachelor of Engineering in Information Technology – RGIT, Mumbai, India

May 2022

TECHNICAL SKILLS

Programming & Scripting: Python, Java, JavaScript, TypeScript, HTML

Backend & APIs: FastAPI, Node.js, REST APIs, Microservices, GraphQL

Cloud & DevOps: AWS (Lambda, DynamoDB, S3), Azure (Functions, Blob Storage), CI/CD, Serverless, Docker, Kubernetes

Databases: MySQL, MongoDB, DynamoDB

Machine Learning & AI: OpenAI GPT, Hugging Face, TensorFlow, YOLOv8, OCR (CRAFT), Pandas

Data Engineering & Analytics: Data Pipelines, Distributed Systems, Power BI, Tableau, Excel

Version Control: Git, GitHub

PROFESSIONAL EXPERIENCE

Minedco Holdings Inc., Leesburg, VA

Jun 2025 – Aug 2025

Software Engineer Intern

- Architected and deployed **backend systems** for an AI-driven storytelling platform, designing a **serverless pipeline** with **Azure Functions**, **Blob Storage**, and the **Facebook Graph API** to ingest and archive millions of posts and images.
- Built scalable **microservices** with **FastAPI** and **async workers** for ingestion, preprocessing, and metadata extraction; deployed via **CI/CD pipelines** on **Azure** and **AWS** for reliable distributed performance.
- Integrated **LLMs** into production pipelines with **embedding-based retrieval (OpenAI + LangChain)**, classification, and summarization services; optimized narrative generation using **caching** and **batching** for low latency and cost efficiency.
- Enhanced system reliability and performance through **exception handling**, **retry logic**, **async task queues**, and **event-driven triggers**, improving throughput by 40% and reducing **API latency** and operational costs.

Cybage Software Pvt. Ltd, Pune, India

Jul 2022 – Jun 2024

Software Engineer

- Designed and implemented **backend services** with **Node.js**, **AWS Lambda**, and **DynamoDB**, improving system performance by 30% and reducing operational costs by 20%.
- Designed and optimized **GraphQL APIs** while enhancing **AWS DynamoDB pipelines**, achieving 98% uptime, secure data handling, and improved query performance across teams.
- Delivered advanced **analytics** and **Power BI dashboards** for **GroupM**, accelerating decision-making by 25% and enhancing resource allocation.
- Conducted **peer code reviews** and enforced **best practices**, achieving 96% **unit test coverage** and reducing **QA feedback cycles** by 30%.

PROJECTS

AI Document Search (RAG Chatbot)

React, FastAPI, LangChain, Pinecone/FAISS, Docker, Vercel

- Built a full-stack “Chat with PDFs” platform enabling **semantic search** and **citation-backed answers** using **Retrieval-Augmented Generation (RAG) pipelines**.
- Developed pipelines for **text extraction**, **embedding generation (OpenAI)**, and **vector search (FAISS/Pinecone)**; deployed frontend on **Vercel** and backend as **Dockerized FastAPI** with **CI/CD**.
- Optimized performance with **streaming responses**, **reranking**, and **caching** (30–50% cost savings), achieving **recall@5 $\geq$ 0.85**, **hallucination rate $<$ 5%**, and **99.9% reliability**.

Facebook Profile Archival & AI Storytelling Platform Python, FastAPI, Azure Functions, Blob Storage, OpenAI GPT, YOLOv8, Stripe

- Engineered a **SaaS platform** to archive complete Facebook profiles (posts, images, metadata) using the **Graph API** with **Azure Functions** and **Blob Storage**, enabling secure, scalable backups with signed URLs.
- Built an AI-powered scrapbook generator by integrating **LLM narratives (OpenAI + LangChain)**, **Dense Captioning**, **OCR (CRAFT)**, and **YOLOv8 object detection**; automated **classification** and **storytelling pipelines** improved content organization and usability.
- Implemented **Stripe Checkout** for seamless \$9.99 monetization, with **production-grade reliability features** including **retry logic**, **error handling**, and session state persistence; reduced downtime risks by 40% during high-traffic loads.